Mathematical Modeling Lecture Notes
Chapter 3 – Model Fitting

I. Model fitting vs. empirical modeling
   a. Now we're going to discuss how to fit functions to a data set.
   b. There are two basic approaches.
      i. Model fitting focuses on an explanation of the data.
         1. For example, we have a set of data, and we want to determine the relationship between independent and dependent variables.
         2. Is the relationship (for example) linear, quadratic, exponential?
         3. With model fitting, the model should make some sense, intuitively.
      ii. Empirical modeling focuses on creating a model that can be used to make predictions.
         1. We'll talk more about this later.
         2. With pure empirical modeling, the model doesn't have to make so much sense, intuitively.

II. Transformations
   a. Sometimes it helps to transform the data when trying to fit a model to it.
      • We've already done this—when we took differences and tried to fit a model to the differences.
   b. A transformation that is often useful is to apply logarithms.
      i. Power relationship, like $y = Ax^p$
         1. We have $\ln y = p \ln x + \ln A$.
         2. So a graph of $\ln y$ vs. $\ln x$ would be linear with slope p and intercept $\ln A$.
      ii. Exponential relationship, like $y = Ce^{kx}$
         1. We have $\ln y = \ln C + kx$.
         2. Thus a graph of $\ln y$ vs. x would be linear with slope k and intercept $\ln C$.
      iii. Logarithmic relationship, like $y = A + B \ln x$
         • A plot of y vs. $\ln x$ would be linear with slope B and intercept A.
      iv. Thus a graph of $\ln y$ vs. x or $\ln y$ vs. $\ln x$ or y vs. $\ln x$ can reveal relationships.
   c. Stopping distance problem, redux.
      i. Show students stopping distance scatterplot again.
      ii. Earlier in the semester we discussed the relationship between speed and stopping distance.
      iii. We thought that either an exponential or (especially) a quadratic model might be a good one.
      iv. Show students scatterplots of $\ln y$ vs. x, $\ln y$ vs. $\ln x$, and y vs. $\ln x$.
      v. The scatterplot of $\ln y$ vs. $\ln x$ looks close to a perfect line with a slope of 2.
   d. Problems with transformations
      i. Transforming the data distorts it.
      ii. When we use an analytic method to construct a model based on the transformed data, it is possible that the distortions will cause the constructed model to be quite inaccurate.
      iii. You always check by comparing the constructed model with the original data.

III. Two common measures of error
   a. In order to select a model that best fits a set of data we need a way to measure the error that a particular model gives.

b. Suppose we have a collection of data points (x_i, y_i), $i \in \{1, \ldots, n\}$ and a function $y = f(x)$ that we are using as a model.
c. The *residual* r_i is the difference between the ith data value and the model's prediction for the ith data value; i.e, $r_i = y_i - f(x_i)$.
d. Chebyshev approximation criterion
   i. The simplest criterion takes as the error $\max |y_i - f(x_i)|$.
   ii. This is called the *Chebyshev approximation criterion.*
   iii. Maybe emphasize that the set of vectors with $\max_i |r_i| = k$ look like n-cubes?
   iv. Let's fit a $y = ax^2$ model to the stopping distance data using the Chebyshev criterion.
   v. One way to construct such a model is to represent it as a linear program.
      1. Let r be the largest value of $|y_i - ax_i^2|$.
      2. Then we want to solve the problem

$$\begin{aligned} &\text{Minimize } r \\ &\text{subject to } |y_i - ax_i^2| \le r \text{ for each } i, i \in \{1, 2, \ldots, n\}; \\ &\qquad\qquad r \ge 0 \end{aligned}$$

      3. Alternatively,

$$\begin{aligned} &\text{Minimize } r \\ &\text{subject to } y_i - ax_i^2 \le r \text{ for each } i, i \in \{1, 2, \ldots, n\}; \\ &\qquad\qquad -y_i + ax_i^2 \le r \text{ for each } i, i \in \{1, 2, \ldots, n\}; \\ &\qquad\qquad r \ge 0 \end{aligned}$$

      4. There are only two variables in this problem as the y_i's and x_i's are known data values.
      5. Thus this is a two-variable linear program and can be solved using the simplex method and software like Excel or Lingo.
   vi. Python can give us the solution to the full model as well.
      1. Show students Python code.
      2. And our model is now $\hat{y} = 1.446x^2$.
      3. Show how this fits the data graphically.
      4. Alternatively, we could have applied the logarithmic transformation.
         - This gives $\ln y_i = \ln a + 2 \ln x_i$.
         - This is linear in $\ln a$, as $\ln y_i$ and $\ln x_i$, for each i, are constants.
         - Following the same procedure (with `[a,r] = fminsearch(@Chebylog,1)`) yields $\hat{y} = 1.915x^2$.
         - Compare this with original data and earlier fit for students.
         - The logarithmic fit doesn't do very well with the larger-valued data points because taking logs compresses those distances by so much. (For example, the difference between $\ln(1.915 * 16^2)$ and $\ln(376)$ is only 0.27.)
e. Minimize sum of absolute deviations
   i. Another approach is to minimize $\dfrac{1}{n}\displaystyle\sum_{i=1}^{n} |y_i - \hat{y}_i|$ (mean absolute deviation) or, more simply, $\displaystyle\sum_{i=1}^{n} |y_i - \hat{y}_i|$ (sum of absolute deviations). Question for students: Why do these give the same model?

ii. Maybe emphasize that the set of vectors with $|r_1| + |r_2| + \cdots + |r_n| = k$ for some fixed k looks like an n-diamond?

iii. Let's fit a $y = ax^2$ model to the stopping distance data using this criterion.

1. So we're trying to minimize $|3 - a| + |6 - 4a| + \cdots + |376 - 256a|$.
2. This is going to be nasty, as the objective function is not differentiable.
3. However, Python can handle it. (Show students.)
4. We get $a = 1.426$, and our model is $\hat{y} = 1.426x^2$ with this criterion.
5. Show how this fits the data graphically.

f. Minimize sum of squared deviations

i. The most common approach, though, is to minimize the *mean squared deviation*, $\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, or, more simply, the sum of squared deviations $\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. (Again, why are these equivalent?)

ii. This measure is more popular for a variety of reasons, one of which is that it is differentiable.

iii. Maybe emphasize that $r_1^2 + r_2^2 + r_3^2 = k$ for some fixed k is a sphere?

iv. Show with Python; the model is $\hat{y} = 1.436x^2$ with this criterion.

IV. Least-squares regression

a. Let's dive a little more deeply into the sum of squares approach.

i. Again, this measure is differentiable.

ii. If we can differentiate the measure with respect to the model parameters then we can apply standard multivariable optimization techniques to find the exact values of the parameters that minimize the sum of the squared differences.

b. Derivation of least-squares formula in the linear case

i. The most basic nonconstant model for a data set is the linear model $\hat{y} = a + bx$.

ii. Let $S = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \sum_{i=1}^{n}(y_i - a - bx_i)^2$.

iii. We have $\dfrac{\partial S}{\partial a} = -2\sum_{i=1}^{n}(y_i - a - bx_i)$ and $\dfrac{\partial S}{\partial b} = -2\sum_{i=1}^{n}(y_i - a - bx_i)x_i$.

iv. Setting these to 0 we have

$$\begin{aligned} an + b\sum_{i=1}^{n} x_i &= \sum_{i=1}^{n} y_i, \\ a\sum_{i=1}^{n} x_i + b\sum_{i=1}^{n} x_i^2 &= \sum_{i=1}^{n} x_i y_i. \end{aligned}$$

v. This is a system of two equations and two unknowns in a and b, with solution

$$\begin{aligned} b &= \frac{n\sum_{i=1}^{n} x_i y_i - \left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}, \\ a &= \frac{\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i\right) - \left(\sum_{i=1}^{n} x_i y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2}. \end{aligned}$$

vi. Does this actually produce a minimum?

1. We have

$$\frac{\partial^2 S}{\partial a^2} = -2\sum_{i=1}^{n}(-1) = 2n \qquad \frac{\partial^2 S}{\partial b \partial a} = 2\sum_{i=1}^{n} x_i$$

$$\frac{\partial^2 S}{\partial a \partial b} = 2\sum_{i=1}^{n} x_i \qquad \frac{\partial^2 S}{\partial b^2} = 2\sum_{i=1}^{n} x_i^2$$

2. Thus

$$d(a,b) = 4\left(n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2\right).$$

3. What is the sign of $d(a,b)$?
   - It doesn't depend on a and b.
   - Does it depend on the x_i's?
4. The sign of $d(a,b)$ will be the same as the sign of the following quantity (divide by $4n^2$): $\dfrac{\sum_{i=1}^{n} x_i^2}{n} - \left(\dfrac{1}{n}\sum_{i=1}^{n} x_i\right)^2$
5. This is $E[X^2] - (E[X])^2$, where X is a random variable having the discrete uniform distribution on $x_1, x_2, \ldots, x_n$.
6. And we know $E[X^2] - (E[X])^2 = Var[X] \geq 0$, with equality holding only when all the data values are the same.
7. Moreover, $\dfrac{\partial^2 S}{\partial a^2} > 0$.
8. Because $d(a,b) > 0$ and $\dfrac{\partial^2 S}{\partial a^2} > 0$ the values of a and b minimize the sum of squared differences.

c. Alternate form of the regression equation
   i. The expressions for a and b can be manipulated into $b = r\dfrac{s_y}{s_x}$ and $a = \bar{y} - b\bar{x}$.
      1. Here, r is the correlation between x and y,
      2. s_y and s_x are the standard deviations of the y and x observations, respectively, and
      3. $\bar{y}$ and $\bar{x}$ are the means of the y and x observations, respectively.
   ii. Among other things, this implies that the regression line $\hat{y} = a + bx$ goes through the point $(\bar{x}, \bar{y})$.

d. Calculating the regression equation on the stopping distances data
   i. For the stopping distance data we do not have a linear model.
   ii. However, the derivation using $\hat{y} = a + bx^2$ is the same with x^2 in place of x (as a and b are the variables here).
   iii. In fact, see text for derivation with any power function $y = Ax^n$.
   iv. Most scientific calculators these days can calculate the regression equation directly, as can Python.
   v. Do calculation for students with Python.
      1. One way is to use the `OLS` function in the statsmodel.api library.
      2. Alternatively, you could have Python calculate the mean, standard deviations, and correlation.
   vi. The regression equation is $\hat{y} = -3.278 + 1.456x^2$.

e. How confident can we be in this calculation?
   i. Correlation
      1. The correlation r gives a measure (also based on the least-squares metric) of the strength of the *linear* relationship between the variables x and y.
         - The closer r to 1 the stronger the positive relationship between x and y.
         - The closer r to -1 the stronger the neg. relationship between x and y.
      2. The square of the correlation, r^2, actually gives the percent of the variation in y that can be explained by a linear relationship with x.
      3. For the stopping distance data $r = 0.9997$, which is really high. (In Python one way to get this is with `np.corrcoef`.)

f. Comparison between least-squares and Chebyshev
   i. Least-squares and Chebyshev are the most commonly-used criteria.
   ii. It turns out there is a relationship between the largest absolute deviation obtained using Chebyshev and the sum of squares obtained via least-squares.
   iii. Let c_{max} be the largest absolute deviation obtained using Chebyshev.
   iv. Let d_{max} be the largest absolute deviation obtained when least-squares is applied.
   v. Clearly, $c_{max} \leq d_{max}$.
   vi. However, if c_i, $i = 1, 2, \ldots, n$, are the deviations obtained using Chebyshev, and d_i, $i = 1, 2, \ldots, n$, are the deviations obtained using least-squares, we have

$$\begin{aligned} & d_1^2 + d_2^2 + \cdots + d_n^2 \leq c_1^2 + c_2^2 + \cdots + c_n^2 \\ \implies \quad & d_1^2 + d_2^2 + \cdots + d_n^2 \leq n c_{max}^2 \\ \implies \quad & \sqrt{\frac{d_1^2 + d_2^2 + \cdots + d_n^2}{n}} \leq c_{max} \leq d_{max}. \end{aligned}$$

   vii. Thus c_{max} is bounded between the square root of the average of the squared deviations when least squares is used and d_{max}.

V. Choosing a best model
   a. There is not a straightforward procedure for deciding between models.
   b. After all, we used different measures to produce our three potential fits for the stopping distance data.
   c. Let's compare the three models with the respective measures.
      i. We have

		Measure		
		Chebyshev	MAD	Avg. Sum of Squares
Method	Chebyshev	5.849	3.253	14.520
	Mean Abs. Deviation	10.935	2.897	14.684
	Sum of Squares	8.319	2.988	13.185
	Linear Regression	6.542	2.430	8.244

      ii. Why did the linear regression perform better than minimizing the mean absolute deviation when using the MAD criterion?
         1. The linear regression has an additional parameter that allows a closer fit to the model.
         2. Closer fitting is not always better, though.
            - The model itself has to make sense.
            - For example, the linear regression model may give smaller errors overall, but it also predicts a negative stopping distance for a speed of 0.

- We'll see more examples of this when we do polynomial interpolation.

iii. Show residual plots as well. There's a clear pattern that shows up in the residual plots of all four graphs.

iv. It's not clear which model is best, though. It depends on how you define "best."

d. Choosing between models

i. Plot the advertising data in the `adData` file. (The data represent the total amount of money spent on advertising in the United States, in millions of dollars, from 1970 to 1989.)

ii. Fit a line to it using Python.

1. This is $y = -401,900 + 5,809.16x$.
2. Plot line and show that it is not a bad fit.
3. We have $r^2 = 0.953$, which is quite good.
4. Unfortunately, there is a pattern to the error.
5. Plotting the residuals shows a clear pattern, too.

iii. Fit an exponential model to it: $y = ae^{bx}$

1. We have $\ln y = \ln a + bx$.
2. Plot $\ln y$ vs. x; it does look roughly linear.
3. Fit a least-squares regression line with $\ln y$ vs x to get $\ln y = 2.43694 + 0.105489x$.
4. Transforming yields $y = 11.43799e^{0.105489x}$.
5. Plot data and residuals; they both look better.
6. The correlation, at $r^2 = 0.993$, is even better.

iv. We also might have a power model; let's check $y = ax^n$.

1. We have $\ln y = \ln a + n \ln x$.
2. Plot $\ln y$ vs. $\ln x$; it does look roughly linear.
3. Fit a least-squares regression line with $\ln y$ vs $\ln x$ to get $\ln y = -25.7094 + 8.35392 \ln x$.
4. Transforming yields $y = e^{-25.7904}x^{8.35392}$.
5. Plot data and residuals; they look similar to the exponential model but perhaps a bit better (especially on the final data point).

v. Comparing the models

1. Data

	Average Sum of Squares
Linear Model	5.525×10^7
Exponential Model	1.536×10^7
Power Model	8.837×10^6

2. Based on the numbers, it appears the power model is the best fit.

Homework. Use Python to find model parameters.

- 3.1 #3, 5
- 3.2 #1, 2a, 2b, 2c. (For #2, use the Chebyshev criterion and the mean absolute deviation criterion.)
- 3.3 #1, 9 (For #9, address the question of which model is the best fit.)
- 3.4 #1, 7ab, 8. For #7 and #8, fit all four models given using least-squares, compute appropriate indicators, and determine which of the four models is best. (One student should get 7ab and 8.)